

Szyborska, 1996: the answer that will not hold



Wisława Szymborska, Kraków, 2009.

Mariusz Kubik, CC BY 3.0

Wisława Szymborska, “The poet and the world”, Nobel Lecture 1996. Polish original; English by Barańczak and Cavanagh. Copyright The Nobel Foundation 1996.

Szyborska, 1996: the answer that will not hold

Stockholm, 7 December 1996. Szyborska is asked to say what a poet does.



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Whatever inspiration is, it's born from a continuous “I don't know.”

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Każdym utworem próbuje na to odpowiedzieć, ale kiedy tylko postawi kropkę, już ogarnia go wahanie ... Więc próbuje jeszcze raz i jeszcze raz.

Each poem marks an effort to answer this statement, but as soon as the final period hits the page, the poet begins to hesitate ... So the poets keep on trying.

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Popper, 1934: conjecture and refutation

Plate not present

`img/popper.jpg`

See `img/IMAGE-SOURCES.txt`

Karl Popper, about 1980.

LSE Library

Karl Popper, *The Logic of Scientific Discovery*, chapter 10, section 85 (1959), the authorised English of *Logik der Forschung* (1934). Located, not yet pinned: a colon and a semicolon version both circulate. See `SOURCES.md`.

Popper, 1934: conjecture and refutation

Vienna, 1934. Popper asks what entitles anyone to believe a general rule drawn from finitely many cases.

His answer is that nothing does, and that this is survivable.

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We do not know: we can only guess.

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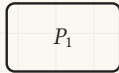
Popper's tetradic schema

The way in which knowledge progresses ... is by **unjustified (and unjustifiable) anticipations**, by guesses, by tentative solutions to our problems, by conjectures. These conjectures are controlled by criticism; that is, **by attempted refutations** ...

Schema after Karl Popper, *Objective Knowledge*, chapter 8 (1972). Quotation: *Conjectures and Refutations*, preface (1963). English original.

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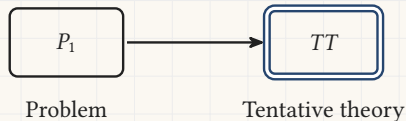


Problem

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Popper's tetradic schema

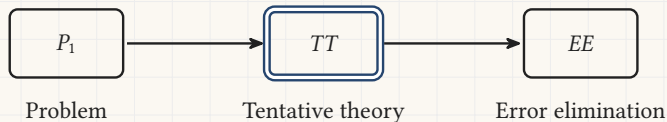
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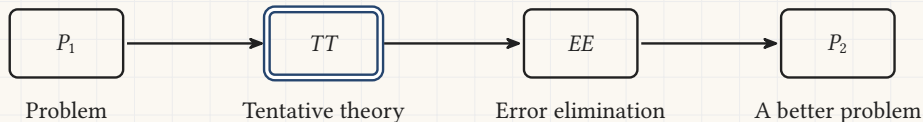
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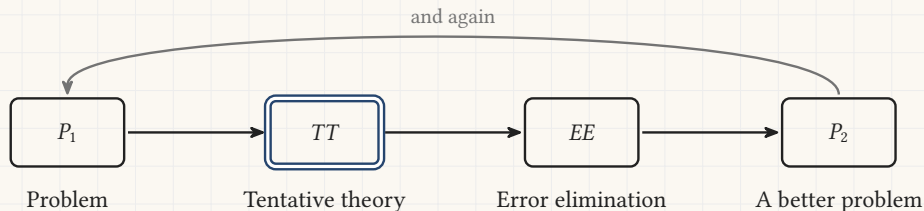
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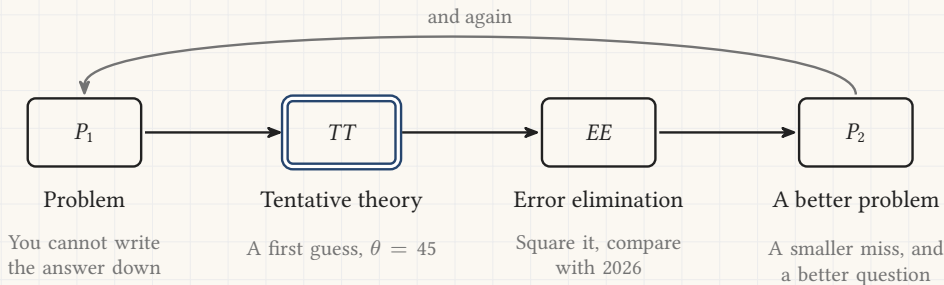
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Astronomy: the recovery of Ceres, 1801

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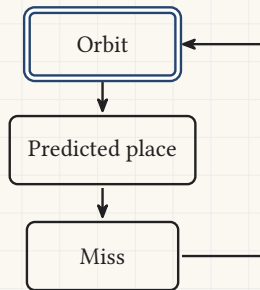
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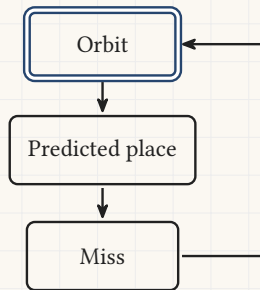
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- Gauss guesses an orbit, predicts, and corrects.
- Von Zach finds it again in December.



Computer science: Turing, 1950



Alan Turing, 29 March 1951.

Elliott and Fry

A. M. Turing, “Computing Machinery and Intelligence”, *Mind* LIX(236), 1950, section 7, Learning Machines. English original; the muted line is this deck’s own reading.

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Elliott and Fry

We cannot expect to find a good child-machine at the first attempt. One must experiment with teaching one such machine and see how well it learns. One can then **try another** and see if it is better or worse.

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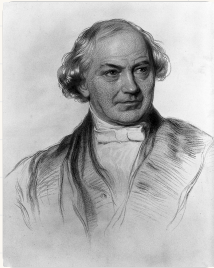
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The loop was written down as the plan for machine intelligence before there was a machine to run it on.

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Whewell, 1858: trying wrong guesses

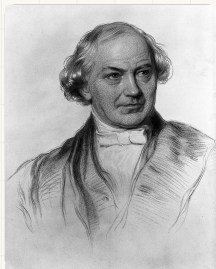


William Whewell, stipple engraving.

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William Whewell, *Novum Organon Renovatum*, third edition (1858), book II, chapter V, section I, page 79. English original; the muted line is this deck's own reading.

Whewell, 1858: trying wrong guesses



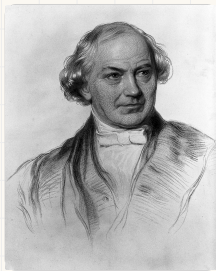
William Whewell, stipple engraving.

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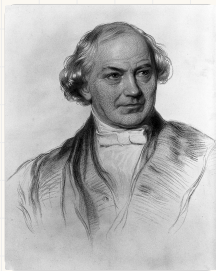
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The tendencies of our speculative nature ... perpetually show their vigour by **overshooting the mark**. They obtain something, by aiming at much more.

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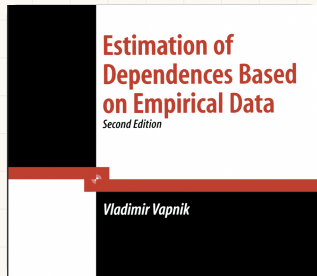
To try **wrong guesses** is, with most persons, the only way to hit upon right ones.

The tendencies of our speculative nature ... perpetually show their vigour by **overshooting the mark**. They obtain something, by aiming at much more.

Most guesses are wrong, and a step that aims too far is how the right one gets found.

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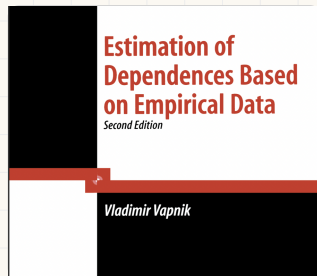
The loop has a name, and the name is a book



Second edition, Springer, 2006

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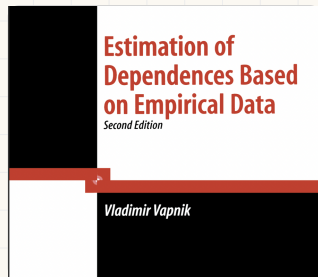


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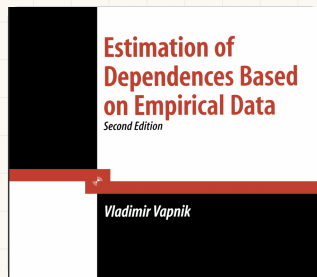
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The book sets out **empirical risk minimisation**, which is the strategy underneath most of what the rest of this course will build.

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The weekend

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Friday 4 September

08:00	Intro and onboarding
09:00	First-order approximation algorithms
10:00	Exercises: first-order optimization
10:30	<i>Coffee, Polysnack</i>
11:00	Gradient descent
12:00	Exercises: gradient descent
13:00	<i>Lunch, Dozentenfoyer</i>
14:00	Neural networks and universal approximation
14:30	Exercises: neural networks
15:30	<i>Coffee, Polysnack</i>

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Saturday 5 September

08:00	Training neural networks with PyTorch
09:00	Exercises: PyTorch
10:00	<i>Coffee, Cafebar</i>
10:30	Validation and overfitting
11:00	Exercises: validation and overfitting
12:00	Project intro: LLM routing

Every lecture is followed by its exercise session. The exercise slugs live on Code Expert and in the onboarding deck.

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16:00	Vapnik's statistical learning theory

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Popper, Karl R. *Conjectures and Refutations*. London: Routledge, 1963. Preface. And *Objective Knowledge*. Oxford: Clarendon, 1972. Chapter 8.

Turing, A. M. “Computing Machinery and Intelligence”. *Mind* LIX(236), 1950, pages 433 to 460. Section 7.

Whewell, William. *Novum Organon Renovatum*. Third edition. London: John W. Parker, 1858. Book II, chapter V.

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